

GARGI WADEKAR ROLL NO.23

```
import numpy as np
import matplotlib.pyplot as plt

# Input range
x = np.linspace(-10, 10, 400)

# 1. Step Function
step = np.where(x >= 0, 1, 0)

# 2. Sigmoid Function
sigmoid = 1 / (1 + np.exp(-x))

# 3. Tanh Function
tanh = np.tanh(x)

# 4. ReLU Function
relu = np.maximum(0, x)

# 5. Leaky ReLU
def leaky_relu(x, alpha=0.2): # Increased alpha for visibility
    return np.where(x > 0, x, alpha * x)

leaky = leaky_relu(x)

# 6. Softmax (line graph)
z = np.linspace(-5, 5, 100)
softmax = np.exp(z) / np.sum(np.exp(z))

# Plot all graphs
plt.figure(figsize=(9, 6))

plt.subplot(2, 3, 1)
plt.plot(x, step)
plt.title("Step Function")
plt.ylim(-0.2, 1.2)

plt.subplot(2, 3, 2)
plt.plot(x, sigmoid)
plt.title("Sigmoid Function")
plt.ylim(0, 1)

plt.subplot(2, 3, 3)
plt.plot(x, tanh)
plt.title("Tanh Function")
plt.ylim(-1.2, 1.2)

plt.subplot(2, 3, 4)
plt.plot(x, relu)
plt.title("ReLU Function")

plt.subplot(2, 3, 5)
plt.plot(x, leaky) # ONE clean line
plt.title("Leaky ReLU Activation Function")

plt.subplot(2, 3, 6)
plt.plot(z, softmax)
plt.title("Softmax Function")
plt.ylim(0, 1)

plt.tight_layout()
plt.show()
```

